1 $H^{(0)}(q \to h(k_1)Z(k_2)\gamma(k_3)$ in 2HDM

$$\mathcal{A} = \mathcal{A}^{\mu\nu}\epsilon_{Z\mu}(k_2)\epsilon_{\gamma\nu}(k_3), \tag{1}$$

$$\mathcal{A}^{\mu\nu} = g^{\mu\nu}\mathcal{A}_{-01} + \sum_{i,j=1}^{3}\mathcal{A}_{ij}k_i^\mu k_j^\nu + \sum_{i,j=1}^{3}\mathcal{B}_{ij}\epsilon^{\mu\nu\alpha\beta}k_{i,\alpha}k_{j,\beta}. \tag{2}$$

Transversed conditions and kinematics,

$$k_2\cdot\epsilon_Z(k_2) = 0, \quad k_3\cdot\epsilon_\gamma(k_3) = 0, \tag{3}$$

$$k_3^2 = 0, \quad k_2^2 = m_Z^2 = y, \quad k_1^2 = m_h^2 = x,$$

$$q_{12} = (k_1+k_2)^2, \quad q_{13} = (k_1+k_3)^2, \quad q_{23} = (k_2+k_3)^2,$$

$$k_3\cdot k_1 = \frac{(q_{13}-m_h^2)}{2} = b, \qquad k_3\cdot k_2 = \frac{q_{32}-m_Z^2}{2} = b, \quad k_1\cdot k_2 = \frac{q_{12}-m_h^2-m_Z^2}{2} = c$$

$$q_{12}+q_{13}+q_{23} = m_h^2+m_H^2+m_Z^2. \tag{4}$$

The photon Ward-identity,

$$k_3^\nu\mathcal{A}_{\mu\nu} = 0. \tag{5}$$

Applying transversed conditions to the amplitude $\mathcal{A}$,

$$\mathcal{A} = \mathcal{A}_{00}\left[\epsilon_Z\cdot\epsilon_\gamma\right] + \sum_{i,j=1}^{3}A_{ij}\left(k_i\cdot\epsilon_Z(k_2)\right)\left(k_j\cdot\epsilon_\gamma(k_3)\right) + \sum_{\substack{i,j=0\\ i\neq j}}^{3}\mathcal{B}_{ij}\epsilon^{\epsilon_Z\epsilon_\gamma\alpha\beta}k_{i,\alpha}k_{j,\beta}. \tag{6}$$

2 CP-even

For CP-even terms (CP-odd contributions vanishes), the nonzero contributions are those with indices $i \neq 2$ and $j \neq 3$,

$$\mathcal{A}^{\mu\nu} = A_{00}g^{\mu\nu} + \sum_{\substack{i,j=1\\ i\neq 2\\ j\neq 3}}^{3}\mathcal{A}_{ij}k_i^\mu k_j^\nu \tag{7}$$

Photon Ward-identity leads to

$$
\begin{aligned}
0 = k_3^\nu \mathcal{A}_{\mu\nu} &= \mathcal{A}_{00} k_3^\mu + \sum_{\substack{i,j=1 \\ i\neq 2 \\ j\neq 3}}^{3} \mathcal{A}_{ij} k_i^\mu (k_3 \cdot k_j) \\
&= k_3^\mu \left(\mathcal{A}_{00} + \sum_{\substack{j=1 \\ j\neq 3}}^{3} \mathcal{A}_{3j} (k_3 \cdot k_j) \right) + \sum_{\substack{i,j=1 \\ i\neq 2,3 \\ j\neq 3}}^{3} \mathcal{A}_{ij} k_i^\mu (k_3 \cdot k_j) \\
&= k_3^\mu \left(\mathcal{A}_{00} + \sum_{\substack{j=1 \\ j\neq 3}}^{3} \mathcal{A}_{3j} (k_3 \cdot k_j) \right) + k_1^\mu \left(\sum_{\substack{j=1 \\ j\neq 3}}^{3} \mathcal{A}_{1j} (k_3 \cdot k_j) \right) \qquad (8)
\end{aligned}
$$

k_i^ν and q^μ are constrained by 4-momentum conservation at each vertices, thus the two, k_3^μ and k_1^μ out of three k_i^μ are free four-vectors. Hence,

$$
\left(\mathcal{A}_{00} + \sum_{\substack{j=1 \\ j\neq 3}}^{3} \mathcal{A}_{3j} (k_3 \cdot k_j) \right) = 0, \qquad \sum_{\substack{j=1 \\ j\neq 3}}^{3} A_{1j} (k_3 \cdot k_j) = 0. \qquad (9)
$$

then,

$$
\mathcal{A}_{00} = -\mathcal{A}_{31}(k_3 \cdot k_1) - \mathcal{A}_{32}(k_3 \cdot k_2), \quad \mathcal{A}_{12} = -\frac{(k_3 \cdot k_1)}{(k_3 \cdot k_2)} \mathcal{A}_{11} \qquad (10)
$$

7 now reads

$$
\begin{aligned}
\mathcal{A}^{\mu\nu} &= \mathcal{A}_{00} g^{\mu\nu} + \mathcal{A}_{11} k_1^\mu k_1^\nu + \mathcal{A}_{12} k_1^\mu k_2^\nu + \mathcal{A}_{31} k_3^\mu k_1^\nu + \mathcal{A}_{32} k_3^\mu k_2^\nu \\
&= \mathcal{A}_{31} \left(-(k_3 \cdot k_1) g^{\mu\nu} + k_3^\mu k_1^\nu \right) + \mathcal{A}_{32} \left(-(k_3 \cdot k_2) g^{\mu\nu} + k_3^\mu k_2^\nu \right) + \mathcal{A}_{11} \left(-\frac{k_3 \cdot k_1}{k_3 \cdot k_2} k_1^\mu k_2^\nu + k_1^\mu k_1^\nu \right) \\
&= \mathcal{A}_{31} P_1^{\mu\nu} + \mathcal{A}_{32} P_2^{\mu\nu} + \frac{\mathcal{A}_{11}}{k_3 \cdot k_2} P_3^{\mu\nu} \qquad (11)
\end{aligned}
$$

where

$$
\begin{aligned}
P_1^{\mu\nu} &= -(k_3 \cdot k_1) g^{\mu\nu} + k_3^\mu k_1^\nu = -a g^{\mu\nu} + k_3^\mu k_1^\nu \\
P_2^{\mu\nu} &= -(k_3 \cdot k_2) g^{\mu\nu} + k_3^\mu k_2^\nu = -b g^{\mu\nu} + k_3^\mu k_2^\nu \\
P_3^{\mu\nu} &= -(k_3 \cdot k_1) k_1^\mu k_2^\nu + (k_3 \cdot k_2) k_1^\mu k_1^\nu = -a k_1^\mu k_2^\nu + b k_1^\mu k_1^\nu. \qquad (12)
\end{aligned}
$$

3 CP-odd

Applying photon WI over the CP-odd terms,

$$\begin{aligned} 0 &= \sum_{i,j=1}^{3} \mathcal{B}_{ij} k_{3,\nu} \epsilon^{\mu\nu\alpha\beta} k_{i,\alpha} k_{j,\beta} = \sum_{i,j=1}^{2} \mathcal{B}_{ij} k_{3,\nu} \epsilon^{\mu\nu\alpha\beta} k_{i,\alpha} k_{j,\beta} \\ &= k_{3,\nu} \epsilon^{\mu\nu\alpha\beta} k_{1,\alpha} k_{2,\beta} (\mathcal{B}_{12} - \mathcal{B}_{21}) \end{aligned} \tag{13}$$

$k_{3,\nu}$ can be abitrary,

$$\mathcal{B}_{12} = \mathcal{B}_{21} \tag{14}$$

Hence,

$$\mathcal{A}^{\mu\nu} = \mathcal{F}_1 P_1^{\mu\nu} + \mathcal{F}_2 P_2^{\mu\nu} + \mathcal{F}_3 P_3^{\mu\nu} + \mathcal{F}_4 \epsilon^{\mu\nu\alpha\beta} k_{1,\alpha} k_{2,\beta} \tag{15}$$

where

$$\mathcal{F}_1 = \mathcal{A}_{31}, \quad \mathcal{F}_2 = \mathcal{A}_{32}, \quad \mathcal{F}_3 = \frac{\mathcal{A}_{11}}{k_3 \cdot k_2}, \quad \mathcal{F}_4 = 2\mathcal{B}_{12}. \tag{16}$$

4 Dual projector

Those $P_{i=1,2,3}^{\mu\nu}$ are not orthogonal basis. Therefore $P_i^{\mu\nu} \mathcal{A}_{\mu\nu} \neq \mathcal{F}_i$. The idea is to find projector such that

$$\mathcal{P}_i^{\mu\nu} \mathcal{A}_{\mu\nu} = \mathcal{F}_i \tag{17}$$

This eliminates the need of tensor reduction for the latter step. The computational form factor now is only a scalar integral. Under polarization summation,

$$\sum_{\text{pol}} |\mathcal{A}|^2 = \mathcal{A}_{\mu\alpha} \mathcal{A}_{\nu\beta} \mathcal{P}_Z^{\mu\nu} \mathcal{P}_\gamma^{\alpha\beta} = \mathcal{F}_i^T G_{im} \mathcal{F}_m \tag{18}$$

with

$$\begin{aligned} \mathcal{P}_\gamma^{\alpha\beta} &= \sum_{\lambda_Z = 0, \pm 1} \epsilon_\gamma^{(\lambda)\alpha} \epsilon_\gamma^{(\lambda)\beta} = -g^{\alpha\beta} + ... \\ \mathcal{P}_Z^{\mu\nu} &= \sum_{\lambda_Z = 0, \pm 1} \epsilon_Z^{(\lambda)\mu} \epsilon_Z^{(\lambda)\nu} = -g^{\mu\nu} + \frac{k_2^\mu k_2^\nu}{m_Z^2} \end{aligned} \tag{19}$$

and G is defined as the Gram matrix,

$$G_{ij} = P_i^{\rho\sigma} P_{Z,\rho}^{\mu} P_{\gamma,\sigma}^{\nu} P_{j,\mu\nu} \tag{20}$$

If G is invertible, define the dual projector

$$\mathcal{P}_i^{\mu\nu} = (G^{-1})_{im} P_m^{\rho\sigma} P_{Z\rho}^{\mu} P_{\gamma\sigma}^{\nu} \tag{21}$$

It satisfies

$$\mathcal{P}_i^{\mu\nu} P_{j,\mu\nu} = (G^{-1})_{im} G_{mj} = \delta_{ij} \tag{22}$$

Therefore,

$$\mathcal{P}_i^{\mu\nu} \mathcal{A}_{\mu\nu} = \mathcal{F}_j \mathcal{P}_i^{\mu\nu} P_{j,\mu\nu} = \mathcal{F}_i. \tag{23}$$

With

$$x = m_h^2, \quad y = m_Z^2 \quad a = (k_3 \cdot k_2), \quad b = (k_3 \cdot k_1), \quad c = (k_2 \cdot k_1), \tag{24}$$

the Gram matrix is,

$$G = \begin{bmatrix} b^2(D-2) & ab(D-2) & -\Delta \\ ab(D-2) & a^2(D-2) - \dfrac{\Delta}{y} & -\dfrac{-c\Delta}{y} \\ -\Delta & -\dfrac{c\Delta}{y} & \Delta\left(x - \dfrac{c^2}{y}\right) \end{bmatrix} \tag{25}$$

where

$$\Delta = a^2 y^2 - 2abc + b^2 x, \quad \det G = -\frac{(D-3)\Delta^3}{y} \tag{26}$$

G is invertible if and only if $\Delta \neq 0$. This contrains the range over $q_{\gamma Z} = q_{23}$ and $q_{h\gamma} = q_{13}$. Its inverse is,

$$G^{-1} = \frac{1}{y \det G} \begin{bmatrix} a^2(D-2)(c^2-xy) + x\Delta & -ab(D-2)(c^2-xy) + c\Delta & a(D-2)(bc-ay) + \Delta \\ -ab(D-2)(c^2-xy) + c\Delta & b^2(D-2)(c^2-xy) + y\Delta & -b(D-2)(bc-ay) \\ a(D-2)(bc-ay) + \Delta & -b(D-2)(ay-bc) & b^2(D-2) \end{bmatrix} \tag{27}$$

and

$$\begin{aligned} \Pi_m^{\mu\nu} &= P_m^{\rho\sigma} P_{Z,\rho}^{\mu} P_{\gamma,\sigma}^{\nu} P_{j,\mu\nu}, \\ \Pi_1^{\mu\nu} &= k_2^{\nu} k_3^{\mu} - b g^{\mu\nu}, \\ \Pi_2^{\mu\nu} &= \frac{k_2^{\mu}(a k_2^{\nu} - b k_1^{\nu})}{y} - a g^{\mu\nu} + k_1^{\nu} k_3^{\mu}, \\ \Pi_3^{\mu\nu} &= \frac{(b k_1^{\nu})(y k_1^{\mu} - c k_2^{\mu})}{y}. \end{aligned} \tag{28}$$

Using (21), the component form of dual projectors, $\mathcal{P}_{i=1,2,3}^{\mu\nu}$ follow.

CP-even and CP-odd are orthogonal, thus its form a block diagonal matrix of the two terms. Changing the projector in CP-even does not alter this property as the Lorentz structure of $\mathcal{A}_{\mu\nu}$ are left untouched. All sanity checks of dual projector over orthogonality among CP-even and with CP-odd is in Dual-Projector.m

5 Decay width

The total decay width,

$$\Gamma_{H\to hZ\gamma} = \frac{1}{256\pi^3 m_H^3}\int_{q_{23}^{\min}}^{q_{23}^{\max}} dq_{23}\int_{q_{13}^{min}}^{q_{13}^{\max}} dq_{13}\sum_{\text{pol}}|\mathcal{A}|^2 \tag{29}$$

where from (18)

$$\sum_{\text{pol}}\|\mathcal{A}\|^2 = 2b^2|F_1|^2 + \frac{a^2y+2abc-b^2x}{y}|F_2|^2 + \Delta\left(x-\frac{c^2}{y}\right)|F_3|^2$$
$$+ 4ab\text{Re}\left(F_1F_2^\star\right) - 2\Delta\text{Re}\left(F_1F_3^\star\right) - \frac{2c\Delta}{y}\text{Re}\left(F_2F_3^\dagger\right) + c^2|F_4|^2 \tag{30}$$

CP-even and CP-odd terms orthogonal to each other under polarization sum.

6 Grouped Diagrams

There is no vertex $HhZ\gamma$ in 2HDM at tree-level, this process is loop-induced. All relevant topologies are listed in (1).

The topologies are grouped as Higgstrahlung topologies with correction $\phi\to Z\gamma$ with $\phi = h^0, H^0$. The Z-radiative off final state topologies with correction $\phi\to h^0\gamma$ with $\phi = A^0, G^0, Z$. The bremsstrahlung topologies with correction $\phi\to h^0 Z$ with $\phi = A^0, G^0, Z$.

There is also the topology group of mixing $Z-\gamma$, self-energies insertion in the photon final legs.

According to [1, 2] because of the existence of coupling ϕZZ[1] where $\phi = h^0, H^0$ and the renormalization of mixing $Z-\gamma$, the mixing counter-terms, $\delta_{Z\gamma}$ are included in the field-renormalization of $Z_{Z\gamma}^{1/2}$. As the result, the vertex counter-terms, $\delta_{\phi Z\gamma}$ cancels out with the mixing counter-terms, $\delta_{Z\gamma}$. It leaves the UV finite summation topologies type 2 to type 9 in Figure 1 with the corresponding topologies in Fig 2.

Among topology types in Fig 1, type 4, 5, 7 and 8, are kinematics forbidden. These topology types contain a vertex of $S\to S\gamma$ which vanishes under the contrains of Lorentz decomposion, transversed conditions (3) and Ward-identity, (5). These vertices do not impose the CP-odd contribution, as such A^0, h^0, γ are CP-conserved. The corresponding topology types of 3, 4,6 and 7 for mixing/vertices counter-terms cancellation of pole propgator, G^0 and A^0 are left vancant.

[1] in [1, 2] the ligh higg, h^0 is used. Here the same argument remains valid for both, H^0 and h^0. They only differs by a factor $\cos(\beta-\alpha)$ or $\sin(\beta-\alpha)$ in couplings H^0ZZ and h^0ZZ.

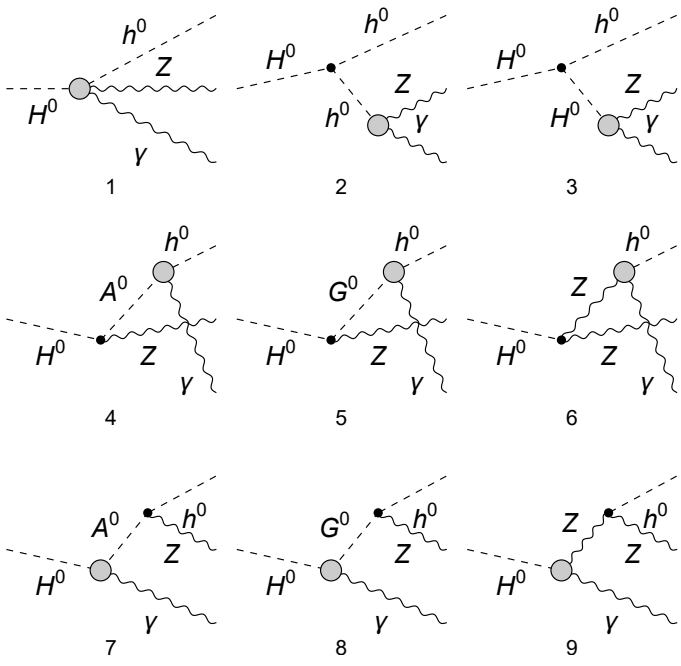


Figure 1: Relevant vertex topologies

For the relevant contributions of type 1 topology in Fig 1, its form factors are calculated in R_ξ gauge (t'Hooft gauge $\xi = 1$). The amplitude can be decomposed as,

$$\mathcal{A} = \mathcal{A}_f + \mathcal{A}_{W,G,u^\pm} + \mathcal{A}_{H^\pm} + \mathcal{A}_{H,G,W,u^\pm} \tag{31}$$

The notation $A_{\text{particles}}$ indicates the internal loop fields contributing to the process.

Higgs and fermion contributions are Gauge independent, while Gauge-boson, Goldstone and Ghost contribution with mixed Higgs contributions are Gauge

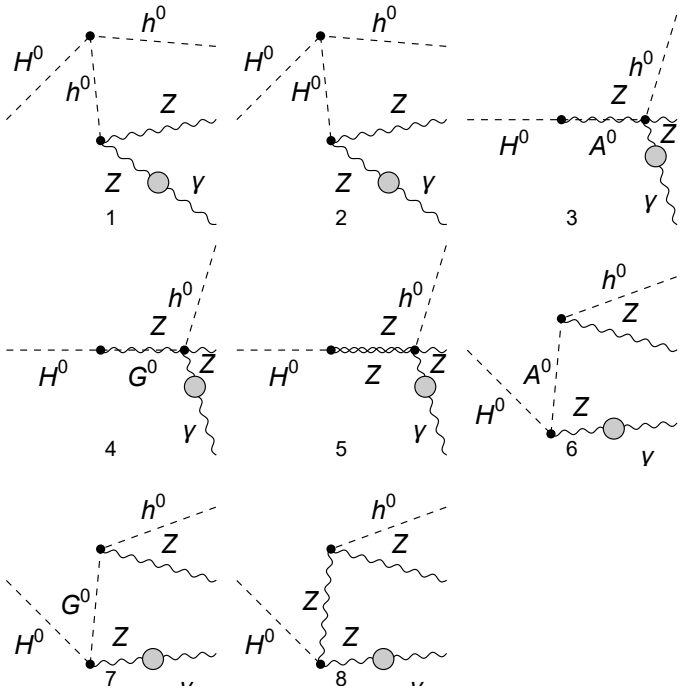


Figure 2: Relevant $Z - \gamma$ self-energies insertion in final photon leg topologies.

dependence. The form factors F_i can be factored further by

$$F_i = F_i^{\text{Box}} + F_i^{\text{Triangle}} + F_i^{\text{Self}} \tag{32}$$

7 Group reduced topologies

8 Higgs loops

The form factor of Higgs Loops,

$$
\begin{aligned}
F_{1,\text{self}} &= \frac{(D-2)}{16\pi^2(D-3)\Delta} g_{H^-H^+H_0h_0} g_{H^+H^-\gamma Z}(ac-bx) B_0(m_{H_0}^2, m_{H^\pm}^2, m_{H^\pm}^2) \\
F_{1,\text{triangle}} &= \frac{i(D-2)(ac-bx)}{8\pi^2(D-3)D\Delta} \Big[D g_{H_0H^+H^-} g_{h_0H^+H^-\gamma Z} C_0(m_{h_0}^2, s_{32}, m_{H_0}^2, m_{H^\pm}^2, m_{H^\pm}^2, m_{H^\pm}^2) \\
&\quad + 4 g_{H^-H^+H_0h_0} g_{H^+H^-\gamma} g_{H^+H^-Z} \Big(B_0(s_{31}, m_{H^\pm}^2, m_{H^\pm}^2) + m_{H_\pm}^2 C_0(m_Z^2, 0, s_{31}, m_{H^\pm}^2, m_{H^\pm}^2, m_{H^\pm}^2) \Big) \Big] \\
F_{1,\text{box}} &= \frac{(D-2)(ac-bx)}{2\pi^2(D-3)D\Delta} g_{H_0H^+H^-} g_{h_0H^+H^-} g_{H^+H^-\gamma} g_{H^+H^-\gamma g_{H^+H^-Z}} \\
&\quad \big[(aD + m_{H^\pm}^2) D_0(m_{h_0}^2, 0, m_Z^2, m_{H_0}^2, s_{32}, s_{31}, m_{H^\pm}^2, m_{H^\pm}^2, m_{H^\pm}^2, m_{H^\pm}^2) + \\
&\quad m_{H^\pm}^2 \big(D_0(m_{h_0}^2, m_Z^2, m_{H_0}^2, 0, s_{12}, s_{31}, m_{H^\pm}^2, m_{H^\pm}^2, m_{H^\pm}^2, m_{H^\pm}^2) + \\
&\quad D_0(m_{h_0}^2, m_Z^2, m_{H_0}^2, 0, s_{12}, s_{32}, m_{H^\pm}^2, m_{H^\pm}^2, m_{H^\pm}^2, m_{H^\pm}^2) \big) + \\
&\quad C_0(m_{h_0}^2, 0, s_{32}, m_{H^\pm}^2, m_{H^\pm}^2, m_{H^\pm}^2) + 2C_0(m_Z^2, 0, s_{31}, m_{H^\pm}^2, m_{H^\pm}^2, m_{H^\pm}^2) \big] .
\end{aligned} \tag{33}
$$

$$
\begin{aligned}
F_{2,\text{self}} &= -F_{1,\text{self}} \\
F_{2,\text{triangle}} &= -F_{2,\text{triangle}} \\
F_{2,\text{box}} &= F_{2,\text{box}}.
\end{aligned} \tag{34}
$$

$$
\begin{aligned}
F_{3,\text{self}} &= \frac{b}{(D-2)(ac-bx)} F_{1,\text{self}} \\
F_{3,\text{triangle}} &= \frac{b}{(D-2)(ac-bx)} F_{1,\text{triangle}} \\
F_{3,\text{box}} &= F_{1,\text{box}}.
\end{aligned} \tag{35}
$$

UV divergences part,

$$
\begin{aligned}
F_{1,\text{self}}^{\text{UV}} &= \frac{(ac-bx)}{4\pi^2(D-4)(D-3)\Delta} g_{H^-H^+H_0h_0} g_{H^+H^-\gamma Z}, \\
F_{1,\text{triangle}}^{\text{UV}} &= \frac{2i(ac-bx)}{\pi^2(D-4)(D-3)D\Delta} g_{H^-H^+H_0h_0} g_{H^+H^-\gamma} h_{H^+H^-Z} \\
F_{1,\text{box}}^{\text{UV}} &= 0.
\end{aligned} \tag{36}
$$

$$
\begin{aligned}
F_{2,\text{self}}^{\text{UV}} &= -F_{1,\text{self}}^{\text{UV}} \\
F_{2,\text{triangle}}^{\text{UV}} &= -F_{2,\text{triangle}}^{\text{UV}} \\
F_{2,\text{box}}^{\text{UV}} &= 0.
\end{aligned} \tag{37}
$$

$$
\begin{aligned}
F^{\mathrm{UV}}_{3,\mathrm{self}} &= \frac{b}{(D-2)(ac-bx)} F^{\mathrm{UV}}_{1,\mathrm{self}} \\
F^{\mathrm{UV}}_{3,\mathrm{triangle}} &= \frac{b}{(D-2)(ac-bx)} F^{\mathrm{UV}}_{2,\mathrm{triangle}} \\
F^{\mathrm{UV}}_{3,\mathrm{box}} &= 0.
\end{aligned} \tag{38}
$$

There is,

$$
\begin{aligned}
g_{H^-H^+\gamma Z} &= -2ig_{H^-H^+\gamma} g_{H^+H^-Z} \\
g_{H^-H^+\gamma Z} &= \frac{ie^2}{s_W c_W}(c_W^2 - s_W^2) \\
g_{H^-H^+\gamma} &= ie \\
g_{H^+H^-Z} &= \frac{ie}{2s_W c_W}(c_W^2 - s_W^2)
\end{aligned} \tag{39}
$$

thus $F_{1,2,3}$ are UV-finite.

9 Fermion Loops

References

[1] A. Barroso, J. Pulido, and J.C. Romão. "Higgs production at $e + e-$ colliders". In: *Nuclear Physics B* 267.3 (1986), pp. 509–530. ISSN: 0550-3213. DOI: https://doi.org/10.1016/0550-3213(86)90128-8. URL: https://www.sciencedirect.com/science/article/pii/0550321386901288.

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